MOTOR: Musculoskeletal Optimisation Through Open Robotics Simulation

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Abstract

Physical AI now estimates and generates moving human bodies at a scale biomechanics has never had, yet the two fields cannot use each other's work: each is fragmented across its own body models, skeletons and solvers, and every bridge between them is a chain built for one body model and one route. We present MOTOR, a single open solver that takes any body physical AI produces, in SOMA's unified form of skin mesh, segment rotations and joint centres, drives the full-body musculoskeletal model MyoFullBody with it, and returns what biomechanics measures: joint angles in the ISB conventions, joint moments, ground reaction forces and the force in every muscle. It runs on physical AI's own GPU stack, Newton and MuJoCo Warp, because the more complex the bodies and movements we simulate, the more parallel a solver has to be. We validate it as a single-camera video pipeline on OpenCap's LabValidation cohort, against OpenCap itself. One solver in place of many chains gives physical AI bodies that obey the mechanics of muscle and bone, and gives biomechanics the scale to study the forces behind human movement far beyond the laboratory.

Gap: MOTOR's headline numbers beside OpenCap's: joint angles, ground forces, joint moments, EMG, time per trial. Waits on the full run (#98).

1 Related work

Physical AI, the branch of artificial intelligence concerned with agents that perceive and act in the physical world, has built a rich set of tools for the moving human body. Video estimators recover a body from a single video [1, 2], motion generators synthesise one from text [3], and GPU simulators such as Newton and MuJoCo Warp [4, 5] run thousands of bodies in parallel. For a long time these tools used many parametric body models, each with its own skeleton [6, 7]. SOMA [8] was made to end that fragmentation. It recovers one unified skeleton from the meshes of SMPL, SMPL-X, MHR and other body models, and estimators and generators such as GEM-X and Kimodo now produce SOMA bodies directly. Physical AI thereby has its first standard body.

Biomechanics has studied the same body for far longer, with the aim of quantifying the forces that produce movement. It records movement with marker-based motion capture and force plates, and musculoskeletal software such as OpenSim [9, 10] estimates from those recordings the joint moments and muscle forces that clinicians and researchers interpret [11]. Its models and solvers are as varied as physical AI's body models once were, and each expects its own inputs [12, 13]. MyoFullBody, a model of the whole body and its muscles from MyoSuite [14], is a candidate to play the role for musculoskeletal models that SOMA plays for body models.

Several works already connect the two fields, but each is built for one body model and one route. OpenCap fits an OpenSim model to keypoints from smartphone video [15, 16]. SKEL re-rigs SMPL with a biomechanical skeleton [17], and HSMR and BioPose regress such poses from images [18, 19].

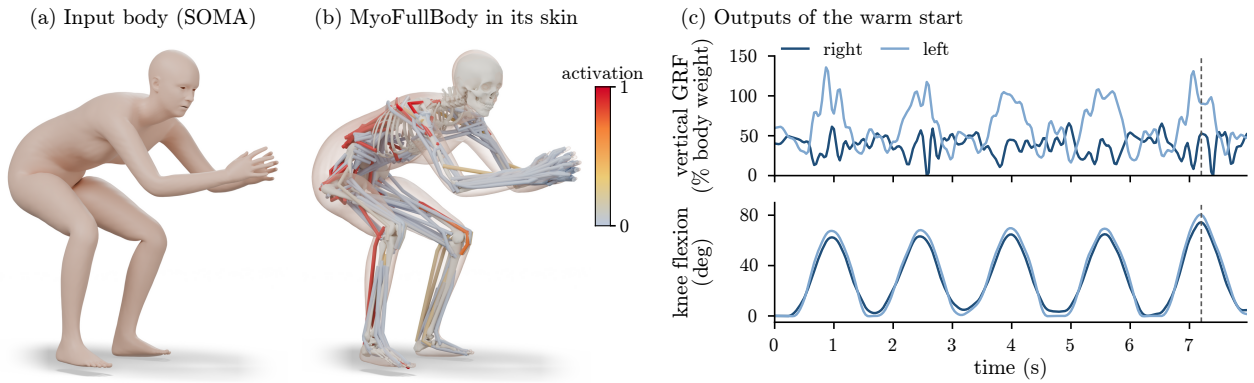


Figure 1: MOTOR on one recording, a squat filmed by LabValidation's 45° camera. (a) The SOMA body GEM-X recovers from the video, at the last repetition's deepest point. (b) MyoFullBody fitted inside that skin, its muscles coloured by activation from 0 (blue-grey) to 1 (red). (c) Each foot's vertical ground reaction force and both knees' flexion; the dashed line marks (a) and (b). The outputs in (b) and (c) are the per-frame chain's, the warm start, not MOTOR's final output (#98).

Cotton and colleagues fit MuJoCo models to video [20–23], and MuscleMimic and MS-Human-700 train muscle-driven bodies on MuJoCo Warp [24, 25]. A body from a new estimator or generator therefore needs a new chain before it reaches biomechanics. What is missing is one solver between the two standards.

MOTOR is that solver. It takes a SOMA body, and through SOMA any body model it unifies, and drives MyoFullBody with it. It returns what biomechanics measures: joint angles in the conventions of the ISB [26, 27] and OpenSim, joint moments, ground reaction forces and muscle forces. It runs on physical AI's own stack, Newton and MuJoCo Warp [4, 5, 28], so that a body is solved where it was made. Unlike the bridges above, MOTOR finds the movement, ground forces and muscle forces of a whole trial in one solve. To our knowledge, its per-frame warm start was already the first open chain from video to muscle forces (fit, residual reduction, static optimisation) on Newton and MuJoCo Warp.

2 Method

MOTOR works in two stages (Figure 3; Appendix A gives the method in full). A calibration, run once and offline, fixes the constants that belong to no recording. A solver then runs on each recording, first fitting MyoFullBody to the moving body and then finding the forces that move it. Both stages rest on the same fit, which we describe first.

MyoFullBody, from MyoSuite [14] with its arm from MoBL-ARMS [29], has 416 muscles and 132 joints. Its bones are sized to the SOMA body's segments, so that no length is left to estimate. Each SOMA segment drives the group of MyoFullBody's bones that covers the same part of the body, through a fixed rest alignment read from the two models' anatomical axes at rest. This alignment is geometry of the two models and is never fitted to motion. The fit then seeks the pose of MyoFullBody closest to the SOMA body within the joints' published ranges. Its main target is the segment rotations, because a rotation fixes a bone's twist about its own long axis, which markers a few centimetres apart resolve poorly. Anatomical landmarks on the skin and the bones place the joint centres, and every bone point outside the skin is penalised, so that the skeleton stays inside the body it came from, an idea SKEL builds into a body model [17].

Calibration, once. The landmarks carry constants that hold for every body, such as each landmark's offset from its bone and its weight in the fit. They are calibrated once, offline, over synthetic SOMA bodies of many shapes and movements, so that they reflect the bodies the solver will meet rather than one study's subjects. A calibration is accepted only when bodies held out of it score within 10% of those used. It is judged on how well the model follows the body, never on any reference joint angle.

The solver, per recording. On a recording only the solver runs. It first fixes the person's own constants, such as where the landmarks lie on their bones, from all of their frames together [23]. It then fits every frame, and this per-frame fit becomes the warm start for one optimisation over the whole trial.

The whole-trial solve is needed because the per-frame fit knows nothing of forces. It represents each joint's path as a smooth spline and keeps the fit's cost over every frame at once. The ground's force follows from the motion itself, through the smooth contact model of OpenCap's tracking simulation [15]. Whatever neither the ground nor the joints explain is left at the pelvis as a residual, which the solve holds under 2% of body weight in force and under the moment limit of Hicks et al. [30]. The muscles are to enter the same solve, so that movement, ground forces and muscle forces are found together and agree with one another. The movement, contact and residual are built, and the muscles' terms are being added (#98).

Why a GPU. For simulation solvers the GPU is a consequence of scale. The more complex the system simulated, the more of its computation must run in parallel, as graphics [31], deep learning [32] and robot simulation [33] have each shown. A full-body model with hundreds of muscles, solved over a whole trial at once, is such a problem. On the GPU, MOTOR also solves many frames and trials side by side, and its differentiable solve offers gradients to the tools around it.

3 Results

We validate MOTOR as a single-camera pipeline, in which GEM-X [1, 34] turns one phone video into a SOMA body and MOTOR solves it. The data are OpenCap's LabValidation cohort [15]: nine healthy subjects filmed by five phones while walking, squatting, rising from a chair and jumping, with motion capture, force plates and EMG as the reference. Each of its 720 single-camera recordings is run on its own. We compare with OpenCap, the only method validated on this cohort and reference, in both its two-camera form [15] and its monocular form [16] (Appendix B).

MOTOR's validation has not yet run. Table 1 and Figure 2 hold its places, and Appendix C those of the full technical validation. The results of the per-frame chain, MOTOR's warm start, are in Appendix D; they are the chain's, not MOTOR's.

4 Discussion

A standard solver between the two fields serves both. For physical AI it means bodies that come with real biomechanics. A motion generator's output, for example, can come back with the ground and muscle forces that would move it, as SCORE already does for the people it generates with Kimodo [3].

For biomechanics it means physical AI's scale. Any estimator whose body SOMA unifies reaches one musculoskeletal model through one solver, so that analyses now confined to laboratory cohorts could extend to large collections of video.

Table 1: MOTOR beside OpenCap on LabValidation. Joint angles on the matched set (45° camera, no drop jumps, the 15 shared angles); the rest over all activities. Ground force in % body weight, both feet summed; moments in % weight × height. –: not reported in our record of that paper.

	OpenCap, 2 cameras	OpenCap Monocular	MOTOR, 1 camera
Joint angle MAE, matched 15 (°)	4.5	4.8	*gap*
Ground force MAE, vert. / ant. / med.-lat.	11.4 / 3.5 / 3.8	–	*gap*
Joint moment MAE, OpenCap's 15	1.2	–	*gap*
Vasti symmetry AUC vs EMG label (squats)	0.83	–	*gap*
Time per trial	44–70 min	–	*gap*

Gap: Figure: (a) MOTOR's vertical ground force per foot against the force plates over the 45° camera's median walking and median squat trial; (b) hip, knee and ankle angle waveforms of the same trials against the lab's inverse kinematics, mean ± 1 SD over cycles. Waits on the whole-trial solver's full run (#98).

Figure 2: MOTOR against the force plates and the lab's joint angles (gap).

Limitations. These claims still wait on the muscle terms and the full run. Only SOMA bodies from GEM-X and Kimodo have yet passed through MOTOR, and a body of another model unified through SOMA, such as the MHR bodies of SAM 3D Body [2], is the test the claim of one input still needs. Errors in the input body pass through to the output, and a single camera leaves depth uncertain. LabValidation contains only healthy adults, and patients whose joints move beyond the published ranges will need ranges of their own.

Estimating the forces behind movement has long required a laboratory and software built for one model and one route. Markerless methods such as OpenCap removed the need for the laboratory. MOTOR aims to remove the bespoke chain by placing one solver between physical AI's standard body and a standard musculoskeletal model. We hope it gives physical AI bodies that obey the mechanics of muscle and bone, and gives biomechanics the scale to study the forces behind human movement far beyond the laboratory.

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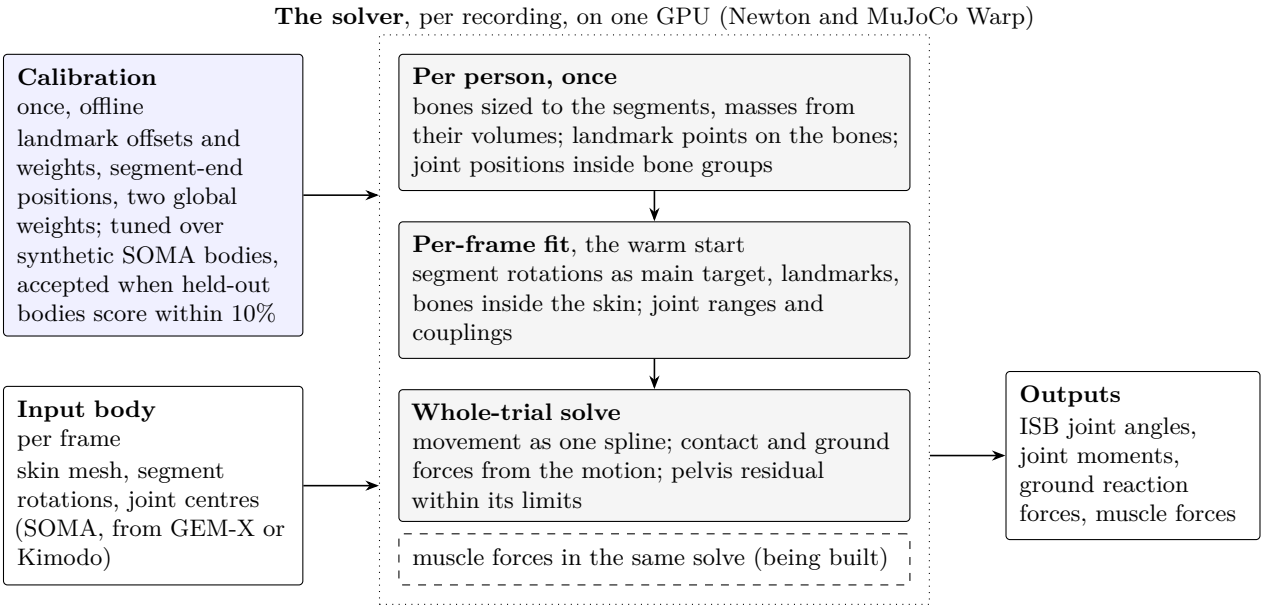


Figure 3: The structure of MOTOR. The calibration runs once, offline, and its constants become part of the solver. On each recording only the solver runs: the person's constants once, the per-frame fit as the warm start, then one solve of the whole trial. The muscles' part of that solve is being built (#98).

A The method in full

This appendix sets out the method in the order in which it runs (Figure 3). The input body and the model are described first, then the correspondence between them and the fit that both later stages share. The calibration, done once and offline, follows, and finally the solver that runs on every recording: the person's constants, the per-frame fit that provides the warm start, and the whole-trial solve. Figure 4 shows the fitted model inside the skin, at full size, in two poses.

A.1 The input body

Any source that provides a skin mesh, segment rotations and joint centres can supply MOTOR's input, although only SOMA bodies, from GEM-X and Kimodo, have so far been run. In the validation the source is GEM-X [1, 34] applied to one smartphone video. GEM-X gives a SOMA body [8] per frame, consisting of its identity (shape), root translation and global segment rotations, at 30 Hz, the rate to which GEM-X resamples every clip. MOTOR takes this body as given and never changes its shape or pose. GEM-X is given the camera's field of view, which MoGe-2 [35] estimates from the same video, and its own contact post-process holds the standing feet still. Because GEM-X's output has no floor, the floor is estimated from the video: a plane is fitted to MoGe-2's points of the floor beside the person, the whole clip is rotated rigidly so that this plane is level, and the clip is lowered until the lowest point of its feet touches it.

A.2 The musculoskeletal model

The model is MyoFullBody from MyoSuite [14], its arm and hand taken from MoBL-ARMS [29, 36]. It has 416 muscles, 86 bones and 132 joints, of which one is the free root, 36 are coupled to another joint by a polynomial (those of the knee and the shoulder girdle, for example) and 95 move freely. Its bone groups (below) are sized to the person's SOMA body, each group's length taken from its SOMA segment, joint to joint, so that MOTOR solves for no length. Its masses are SOMA's

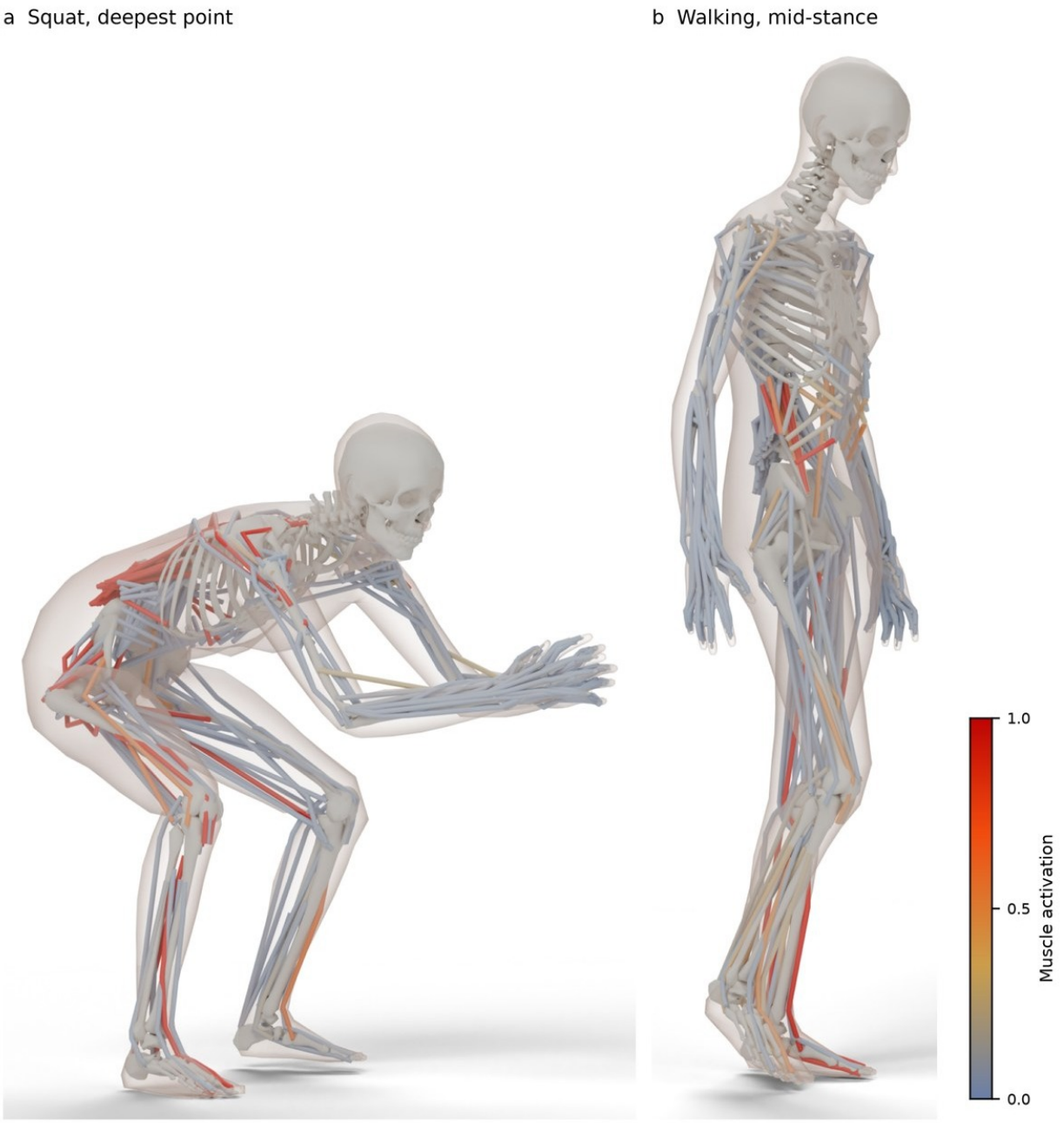


Figure 4: MyoFullBody fitted inside a person's skin from one LabValidation video: (a) a squat at its deepest point, (b) walking at mid-stance. Drawn from the saved results of MOTOR's earlier per-frame chain, its warm start; the muscles are coloured by that chain's activations, not the whole-trial solve's.

segment volumes distributed over the subject's measured mass.

A.3 Segments, bone groups and landmarks

Segment rotations and bone groups. Each SOMA segment is coupled to a *bone group*, the bones of MyoFullBody that cover the same part of the body, one of which, the group's *frame bone*, carries the segment's rotation (Appendix E). A spine segment is coupled to the vertebrae at its height, paired by their share of the spine's height, and the hand to the bones of the wrist and hand, with the lunate carrying its rotation. Each segment drives one group. SOMA has two neck segments where MyoFullBody has a single rigid neck; the first drives it, and the second drives nothing, although its rotation error is still reported.

How a segment's frame lines up with its group's frame is read once from the two models at rest (SOMA's template in its bind pose, MyoFullBody in its default pose with its couplings applied) and kept as a fixed *rest alignment* per group. Each group's anatomical axes are constructed on both models from the same anatomy: the line between the group's end joints or between landmarks, the bone's own axis, or a hinge line, such as the line through the first and fifth metatarsal heads that the foot and toes share. The feet are then turned about that hinge line until SOMA's foot, flat on its floor at rest, corresponds to MyoFullBody's heel bone flat in its default pose, and the groups whose bones would stand outside SOMA's skin along those lines (the head first) are turned and shifted rigidly at rest to their least depth outside it. The rest alignment is therefore geometry of the two models: it is never fitted to motion data or to any reference joint angle, and never per person or per clip.

Virtual markers on anatomical landmarks. Beside the rotations, the fit uses virtual markers on anatomical landmarks, each defined by its anatomy following the ISB recommendations [26, 27]: the acromion, the humeral epicondyles, the anterior and posterior superior iliac spines, the malleoli, the metatarsal heads and others. Each landmark sits on a fixed point of the SOMA mesh, given as a face and barycentric weights, so that it moves with the person's skin, and on the matching point of a MyoFullBody bone, plus an offset from the bone toward the skin. A study of the two models placed 102 landmarks, of which the fit uses 84, each with its own weight. No licensed dataset's marker list is used.

Joint centres as segment ends. SOMA's rig joints mark where each bone group begins and ends: the hips, knees, ankles, shoulders, elbows, wrists and the joints between spine segments. Each such end acts as one more marker, placed at MyoFullBody's own joint centre there and aimed at SOMA's joint. The joints inside a group (the lumbar levels, the shoulder blade, the wrist bones and the joints of the inner foot) are aimed at nothing; the landmarks and the model's couplings place them. No joint is treated as a special case.

The skin. The bones belong inside the body from which they came. SKEL builds this idea into a body model, a biomechanical skeleton placed inside the skin of SMPL [17]; MOTOR instead uses the input's own skin as a target. Each frame's SOMA skin is posed, and a fixed subset of the vertices of every bone mesh is measured against it: each vertex's offset to the closest point of the skin, or zero when the vertex lies inside, as decided by the generalised winding number [37].

A.4 The fit

The fit is the objective that the calibration, the per-frame warm start and the whole-trial solve share. For one frame, with the person's constants fixed, it finds MyoFullBody's coordinates q that come closest to the input body while every joint stays within its anatomical limits:

$$\min_{q} \frac{1}{N_g}\sum_{g}\left\|\log\!\left(\hat{R}_g^{\top} R_g(q)\right)\right\|^2 \;+\; \frac{\lambda_m}{\sum_m w_m}\sum_{m} w_m\left\|p_m(q)-\hat{x}_m\right\|^2 \;+\; \frac{\lambda_s}{N_b}\sum_{b}\left\|d_b(q)\right\|^2 \qquad (1)$$
$$\text{s.t. } q_{\text{follower}} = f(q_{\text{leader}}) \text{ for every coupling}, \qquad q_{\min} \le q_{\text{leader}} \le q_{\max}.$$

The first term is the main one: for each of the N_g bone groups, it measures the angle between the frame bone's orientation $R_g(q)$ and its SOMA segment's rotation $\hat{R}_g$ composed with the group's rest alignment. The second pulls each marker m (the landmarks and the segment ends) at its point $p_m(q)$ on the model toward its point $\hat{x}_m$ on the SOMA body, weighted by w_m. The third holds the N_b bone vertices inside the skin, d_b being a vertex's offset outside it. Each goal is normalised by its own count, as OpenSim's inverse kinematics weighs its goals [10]. The markers' goal λ_m is set by the calibration (2.1169), and the skin goal λ_s is 10^3, chosen in a sweep that traded the bones outside the skin against the rotation error. A coupled joint (a *follower*) moves with a *leading* coordinate through MyoFullBody's own polynomial f, held as an equality constraint, and the limits are the published ranges of the leading coordinates, held as bounds. The fingers are held at SOMA's rest.

The reference solver is mink [38], a differential inverse-kinematics library on MuJoCo. Each of its steps solves one quadratic programme, in which the segment rotations and the marker positions are tasks on frames of the model, the couplings are equality constraints, the leading coordinates' ranges are limits and the finger joints are frozen; the step length is controlled as in Levenberg–Marquardt. The skin term adds three rows per bone vertex, its offset from the closest point of the skin, whose Jacobian keeps only the motion away from the closest face, edge or vertex of the skin mesh. On the GPU the same step is a damped Gauss–Newton step in the leading coordinates, solved for every frame at once with Warp's tile Cholesky factorisation [28]. On every frame of a development clip it matches mink within 0.0025° in every joint angle and 0.0042 mm in every model point. A frame that does not converge in 300 steps stops the run with an error rather than being kept silently.

A.5 Calibration, once and offline

The calibration (level 1 in the code) determines the constants of the landmarks that hold for every body, and it runs once, offline, before the solver is applied to any recording. One bounded least-squares problem tunes, all together, each landmark's offset from its bone (for a landmark that defines a joint, only along its bone); each landmark's weight, mirrored and shared between left and right; each segment end's position on MyoFullBody, within 40 mm of the model's own joint centre; and two global weights, the markers' goal λ_m and the segment ends' weight among the markers. Its objective has two terms, each normalised by its value at the start so that the two weigh alike: each body region's bones' depth outside the SOMA skin, and each segment's rotation error against its bone group. Each evaluation runs the person stage and the per-frame fit on every tuning body.

The bodies are synthetic SOMA bodies (tall, short, heavy, slim, long- and short-legged), each performing seven clips (a Kimodo walk, a rig walk, arm raises, a reach, a squat, running and sit-to-stand). They are split by body into tuning and check bodies, with every clip of a body on one side. The check bodies are scored during the run but never used to choose, and a tuned point is accepted only when they score within 10% of the tuning bodies. The calibration's gradient is taken through the person stage and the per-frame fit by implicit differentiation; the columns that path does not yet cover are taken by finite differences, and every column is checked against finite differences before

a run is used. The calibration thus sees only synthetic bodies' skin depth and rotation error, never any reference joint angle.

A.6 The solver, per recording

The person's constants (level 2). With the calibration fixed, the person's constants are solved from all of their frames together [23]. About 30 frames, spread over the person's clips in proportion to their length, are posed, and one sparse least-squares problem solves, together with those frames' coordinates, each landmark's point on its bone and the position of each joint inside a bone group, each within 3 cm of where the calibration and the model placed it and drawn back toward it by a weight. Only what the data can distinguish is solved: a joint position whose effect on the residuals is a combination of the other unknowns' effects stays where the model has it. The landmarks on the soles' pads and on the skull never move, so that a foot cannot sink through the sole of the skin nor the skull through the head. This stage has no skin term, because with one the solve moved the inside joints by 40 to 52 mm to reduce the bones' depth outside the skin instead of fitting the person. The constants are written into the person's model file, so that every later solve uses them.

The per-frame fit, the warm start (level 3). With the person's constants fixed, every frame solves only its pose, by Equation (1): the rotations as the main term, the markers beside them, the bones inside the skin, and the joint limits and couplings as hard constraints. The resulting path is the whole-trial solve's starting point.

A.6.1 The whole-trial solve

The per-frame fit follows the body but knows nothing of forces: it can place a foot where no ground could hold it, and the frames it gives need not be consistent with the forces that would move the model between them. MOTOR therefore solves the whole trial as one optimisation, with the per-frame fit as its starting point (its *warm start*). The movement, the dynamics with contact and residual, and the contact's physical settings are built and described here; the muscles, batching across trials and the solve's use inside SCORE are being built.

The movement. Every leading coordinate's path over the trial is one uniform cubic B-spline with one control point per frame, so that speeds and accelerations are the spline's own derivatives and no kinematic defects need be constrained. The root's position is a spline like a coordinate's. Its orientation is the per-frame fit's orientation, low-pass filtered, times a correction rotation whose rotation vector, in the body frame, is again a spline. The movement's cost is the fit's own cost of Equation (1) over every frame at once, plus the spline's jerk, with the leading coordinates kept inside their ranges by bounds on their control points. The jerk weight of each coordinate is not tuned: it is the weight that places the solve's smoothing at the 6 Hz cut-off with which the per-frame chain filters [11]. Each frame's Gauss–Newton block is assembled from the GPU fit's own pieces, and the normal equations are banded, since each frame's rows touch only its three nearest control points.

Dynamics, contact and the residual. The *dynamics solve* adds the dynamics of every frame to the same unknowns. MuJoCo Warp, inside Newton, gives the generalised forces $\tau_{\text{needed}}(q, \dot{q}, \ddot{q})$ that each frame's motion requires. The ground's force follows from the motion alone, with no force unknowns: every point of the person's skin within 2 cm of its floor is pushed by a smooth spring

and damper of the Hunt–Crossley kind with a smoothed Coulomb friction, the form of OpenCap's tracking simulation [15],

$$f_n = k\,\sigma_s(-h)\,\sigma\big(1 - c\,v_n\big), \qquad \mathbf{f}_t = -\mu\,f_n\,\frac{\mathbf{v}_t}{\sqrt{\|\mathbf{v}_t\|^2 + \varepsilon^2}}, \tag{2}$$

with h the point's height over the floor, v_n and $\mathbf{v}_t$ its normal and tangential speed, k the stiffness, c the dissipation, μ the friction, and σ_s and σ smoothed ramps, so that a lifting foot pushes nothing and never pulls. The forces left for the body to supply, on the leading coordinates, are

$$\tau = G^\top\big(\tau_{\text{needed}} - J_c^\top f_c\big), \tag{3}$$

where G maps the leading coordinates' velocities to those of all coordinates (the couplings' Jacobian), f_c stacks the contact points' forces and J_c is the Jacobian of their positions. The root's six rows of τ are the pelvis residual, the force and moment that neither the body nor the ground explains. An augmented Lagrangian holds them on every frame under 2% of body weight in force and 1% of body weight times the centre of mass height in moment; the moment limit is that of [30], and the force limit is our own, tightened from 5% with AddBiomechanics' results [13] as the benchmark. The remaining rows are the joints' torques, kept small as an effort term.

The whole is solved by Levenberg–Marquardt on the banded normal equations. The Jacobians of the needed forces come from three sources: MuJoCo's mass matrix for the accelerations (exact), the joints' damping for the speeds, and one-sided finite differences on the GPU for the positions. Before the first iteration the contact is settled: where a frame's lowest skin point lies under the floor at the warm start, that frame's floor is lowered to it. Without this step, a seat whose height is taken from the protocol, while the video places the person's skin lower, pushed the body with many times its weight.

Settings. The contact's settings are physical: a whole foot's stiffness of 6×10^6 N/m spread over its skin points (OpenCap's six contact spheres per foot at 10^6 N/m each), a dissipation of 2.0 s/m and a friction of 0.8 as in OpenCap's contact spheres, and a slip transition ε of 0.05 m/s. The effort term's weight is set so that, at the warm start, it amounts to 0.1 of the fit's cost. These settings were tuned on subjects 2 to 5 only, against the force plates with the dynamics solve's own contact, and the tuning kept these first values. Nothing is ever fitted to the reference joint angles.

Muscles. The muscles are to enter the same solve through per-person polynomial muscle–tendon lengths, whose derivatives are the moment arms, fitted to MuJoCo's own tendon lengths, together with MuJoCo's muscle force curves, so that the activations are solved jointly with the motion and the contact rather than distributed afterwards.

> **Gap:** The muscles' formulation as built (the muscle terms, the activation dynamics and how they enter the solve) replaces this box once it is built and measured (#98).

B Validation protocol in full

Camera input. The video comes from five synchronised iPhones, Cam0 to Cam4 in LabValidation's numbering. The 144 non-static trials, each seen by five cameras, give 720 single-camera inputs, every one run on its own. For the matched comparison we use the 45° camera (Cam1) on the non-jumping activities and the fifteen joint angles that have a counterpart in OpenCap Monocular's

set (Appendix B), the envelope on which OpenCap Monocular reports its results [16] and on which the author's earlier work reported GEM-X's angles [39]. The full technical validation pools all 720 inputs, every activity and every camera.

Metrics. Because video and reference are not synchronised at the millisecond level, each trial receives one time shift of the reference before scoring. Joint angles are scored by mean absolute error (MAE), centred MAE (cMAE, with each trial's constant offset removed, which separates tracking from offset), root mean square error and Pearson's r, with a 95% confidence interval from a per-subject bootstrap of 20,000 draws. Ground reaction forces are scored as OpenCap scores them, in % of body weight on three axes with both feet summed, and per foot by the correlation of the vertical force. Joint moments are scored in % of weight times height over OpenCap's fifteen, with the knee adduction moment's early-stance peak reported separately. In squats, the vasti symmetry index is scored by its AUC against the EMG's own asymmetry label. Activations are scored by on–off agreement and correlation with each EMG channel, and the pelvis residual by the share of frames within both limits.

Time. The solve time is MOTOR's wall time per trial on one named GPU, from the input body to its output, set beside OpenCap's recorded solves in its LabValidation release, whose hardware is not recorded.

Joint definitions. The matched comparison scores fifteen joint angles, the coordinates that have a counterpart in OpenCap Monocular's set: ankle angle, knee angle, hip flexion, hip adduction and hip rotation on both sides, together with pelvis tilt, pelvis list, lumbar extension, lumbar bending and lumbar rotation. The full technical validation also scores the shoulders (arm flexion, adduction and rotation), the elbows, the forearms' pronation–supination, the subtalar and metatarsophalangeal angles and pelvis rotation (Appendix D). Each angle is read from MyoFullBody's own coordinates after the fit, and nothing is calibrated or regressed against the reference. The hip, knee, ankle, subtalar, toe and elbow angles are MyoFullBody's own coordinates of the same name. The pelvis and lumbar angles are OpenSim's intrinsic Z, X, Y angles of the model's own body frames, as the lab's model defines them, and the shoulders are the same angles of each humerus on the torso. Pronation–supination has its zero at a different forearm pose in each model, so it is scored only by cMAE and r. The scoring follows the author's earlier work [39]: one sub-frame time shift of the reference per trial, found on the centred residual of both sides' hip, knee and ankle flexion; MAE, cMAE, RMSE and r; a dagger where the reference moves less than 15°; and the per-subject bootstrap of 20,000 draws.

C Full technical validation of MOTOR

This appendix follows the layout of Appendix B of [39]. It covers all 720 single-camera inputs, with every activity including the drop jumps and each camera as its own input, and reports every joint angle MOTOR outputs, the ground reaction forces on all three axes per activity, the joint moments, the residuals and every EMG channel, per activity and per camera. Every table waits on the whole-trial solver's full run (#98); Appendix D gives the same tables for the warm start.

C.1 Every joint angle

Table 2: MAE, cMAE, RMSE and r per coordinate on the envelope and on all inputs, the matched 15 first.

Gap: Every coordinate MOTOR outputs (the 31 of Appendix D's table): MAE, cMAE, RMSE and mean r on the matched envelope and over all scored inputs, the dagger for a reference range under 15°, and the pooled matched 15; the warm start beside each.

Table 3: Per coordinate and activity.

Gap: MAE and cMAE of every coordinate per activity (walking, walking with trunk sway, squats, sit-to-stand, drop jump), with cycle-aligned waveforms of the hip, knee and ankle per activity.

C.2 Per subject group, activity and camera

Table 4: Per group: inputs, matched MAE and cMAE, vertical GRF MAE, median vertical r per foot, joint moment MAE, mean EMG r.

Gap: Rows: each subject, each of the five activities and each of the five cameras (Cam0 to Cam4); MOTOR's values with the warm start's in brackets.

C.3 Ground reaction force

Table 5: Ground reaction force per activity on all three axes.

Gap: MAE in % body weight, vertical, anterior and medio-lateral, both feet summed as OpenCap scores them, per activity; the share of the plates' stance frames each foot carries; the centre-of-pressure error; OpenCap's two-camera row beside it.

C.4 Joint moments

Table 6: Joint moments per coordinate and activity.

Gap: MAE in % weight × height for each of OpenCap's fifteen joint moments per activity, the knee adduction moment's early- and late-stance peaks in walking, against OpenSim's inverse dynamics on the lab's model; OpenCap's published values beside them.

C.5 Residuals

Table 7: The pelvis residual per group.

Gap: Per group: the median force and moment, the 95th-percentile force, and the share of frames within both limits (2% body weight, 1% weight × centre of mass height).

C.6 Muscle activations against EMG

Table 8: Each of the 16 EMG channels against MOTOR's activation of that muscle.

Gap: On–off agreement, Pearson's r and the lag at the cross-correlation peak for each channel (biceps femoris long head, semitendinosus, gastrocnemius medialis, soleus, gluteus medius, vastus lateralis and medialis on both sides, rectus femoris and tibialis anterior on the left), per activity; the vasti symmetry AUC per camera against the instructed and the EMG-labelled asymmetry.

C.7 Time per step and repeatability

Table 9: MOTOR's wall time per step on one GPU.

Gap: Median, range and per-frame seconds of each step (fit levels 2 and 3, the whole-trial solve's movement, dynamics and muscles) over all scored inputs, the GPU named; two runs compared bit for bit.

C.8 Failures

Table 10: The inputs without a score, by cause.

Gap: Every input of the 720 that did not run or was not scored, by cause.

D The per-frame chain, MOTOR's warm start

Everything in this appendix is the per-frame chain's, measured on 7 November 2026; none of it is MOTOR's result. Before the whole-trial solve existed, MOTOR's forces came from a chain of per-frame steps, which now gives the whole solve its starting point. After the fit, the standing feet are put on the floor by moving the whole body up or down, never by changing the pose. A residual reduction in the manner of OpenSim's RRA then adjusts the pelvis, the trunk and the masses within a per-frame cap, so that the ground can explain the movement. Last, the forces are split in the two stages of OpenSim's static optimisation [9]: the ground reaction force from the pelvis' six balance equations alone, then the muscle activations with that force fixed. Every step runs every frame at once on the GPU in Warp's deterministic mode, and two runs agree bit for bit, which answers, for this chain, the CPU against GPU drift reported on MyoSuite bodies [40].

In the chain's final run all 720 single-camera inputs were run; 715 were scored, and Section D.8 gives the rest by cause. "This run" below is that run; "before" and "the previous full run" are the chain's earlier full run of the same 720 inputs (700 scored), before changes to its contact model, its floor and its joints' ligaments, scored again with the same code. Every number is from paper/motor/evidence/labval-final-2026-11-07.txt, except the published solve times of Table 18 and where a comment names another source. Some of the restored figures' legends say "MOTOR": they were drawn when the chain carried that name, and show the chain. On the matched set the chain's joint angle error is 5.16° (95% CI 4.77 to 5.59°), its vertical ground reaction force error 16.2% of body weight and its joint moment error 2.41% of weight times height; it takes a median of 66 s per trial on one NVIDIA L4 GPU, and its pelvis residual stays within both limits on only 21% of frames.

D.1 Every joint angle

Table 11: MAE and centred MAE (degrees) per coordinate, previous full run → this run, and this run's mean r (†: the reference moves less than 15°). Envelope: 45° camera, no drop jumps (72 → 89 trials). All: every scored input (700 → 715). The matched 15 first; pro_sup has no shared zero, so only its cMAE.

Coordinate	Envelope MAE	Envelope cMAE	Envelope r	All MAE	All cMAE	All r
ankle angle r	6.41 → 6.39	3.18 → 3.19	0.89	5.97 → 5.96	3.94 → 3.92	0.92
ankle angle l	4.45 → 4.61	3.22 → 3.32	0.89	5.52 → 5.51	3.75 → 3.75	0.92
knee angle r	6.30 → 6.06	3.16 → 3.04	0.99	6.02 → 5.99	3.18 → 3.17	0.99
knee angle l	6.10 → 5.94	3.33 → 3.24	0.98	6.11 → 6.09	3.51 → 3.50	0.98
hip flexion r	8.17 → 8.03	3.90 → 3.82	0.98	8.91 → 8.89	4.16 → 4.11	0.98
hip flexion l	7.46 → 7.37	3.56 → 3.53	0.98	8.96 → 8.93	4.22 → 4.17	0.98
hip adduction r	4.20 → 4.20	2.81 → 2.73	0.52†	3.69 → 3.72	2.58 → 2.59	0.52†
hip adduction l	4.78 → 4.72	2.83 → 2.76	0.60†	3.48 → 3.52	2.50 → 2.52	0.57†
hip rotation r	4.10 → 4.07	2.62 → 2.62	0.43†	4.38 → 4.36	2.77 → 2.77	0.51
hip rotation l	4.53 → 4.41	3.19 → 3.06	0.43†	4.22 → 4.20	3.03 → 3.00	0.54
pelvis tilt	5.93 → 6.06	3.60 → 3.64	0.51	7.05 → 7.02	4.43 → 4.39	0.40
pelvis list	3.26 → 3.15	2.48 → 2.46	0.17†	3.10 → 3.11	2.22 → 2.24	0.16†
lumbar extension	4.66 → 4.56	3.15 → 3.11	0.64	5.36 → 5.32	3.37 → 3.35	0.75
lumbar bending	4.37 → 4.19	3.31 → 3.21	0.44	3.68 → 3.69	2.51 → 2.54	0.45†
lumbar rotation	3.67 → 3.67	2.14 → 2.12	0.58†	4.01 → 3.97	2.74 → 2.71	0.44
arm add l	3.79 → 3.98	2.33 → 2.34	0.70	6.62 → 6.58	5.18 → 5.12	0.70
arm add r	4.30 → 4.23	2.25 → 2.23	0.51†	6.22 → 6.16	4.45 → 4.40	0.64
arm flex l	7.92 → 8.03	3.11 → 3.19	0.93	12.59 → 12.46	8.72 → 8.59	0.90
arm flex r	6.77 → 7.00	3.09 → 3.19	0.93	11.32 → 11.20	8.65 → 8.52	0.90
arm rot l	8.19 → 8.01	4.41 → 4.50	0.64	12.48 → 12.35	9.91 → 9.78	0.67
arm rot r	7.25 → 7.03	4.27 → 4.09	0.67	11.31 → 11.18	8.74 → 8.61	0.68
elbow flex l	23.80 → 23.68	4.74 → 4.61	0.74	24.77 → 24.70	5.73 → 5.67	0.79
elbow flex r	20.96 → 21.12	3.07 → 3.00	0.80	22.79 → 22.74	5.84 → 5.76	0.76
mtp angle l	15.23 → 15.35	2.49 → 2.47	0.15†	15.73 → 15.74	2.19 → 2.19	0.10†
mtp angle r	15.17 → 15.22	2.58 → 2.55	-0.02†	14.99 → 14.99	2.05 → 2.06	0.00†
pelvis rotation	2.76 → 2.70	2.63 → 2.57	0.50†	2.66 → 2.63	2.45 → 2.42	0.48†
pro sup l	-	2.82 → 2.82	0.16†	-	3.63 → 3.61	0.25
pro sup r	-	2.44 → 2.55	0.18†	-	3.48 → 3.46	0.26†
subtalar angle l	9.18 → 9.24	7.02 → 6.88	-0.21	9.74 → 9.73	6.25 → 6.25	-0.07
subtalar angle r	9.02 → 9.16	6.26 → 6.26	0.04	11.44 → 11.39	6.21 → 6.21	0.04
matched 15, pooled	5.23 → 5.16	3.10 → 3.06		5.36 → 5.35	3.26 → 3.25	

The angles are the same chain's in both runs (GEM-X and the fit are unchanged); the small differences come from the inputs each run has. The elbows are about 21 to 25° off with a centred error of 3 to 6° (straighter than the lab's in each of the 89 envelope trials; GEM-X's arm estimate, which the fit follows), and the toes (mtp) 15° off with a centred 2 to 3°: constant offsets, not tracking errors.

D.2 Per subject group, activity and camera

Table 12: This run per group (the previous full run's value in brackets): inputs, matched MAE and cMAE (degrees), vertical GRF MAE (% body weight, both feet), median vertical r per foot, joint moment MAE over OpenCap's 15 (% weight × height), mean EMG r over the channels. Cameras: LabValidation's numbering; Cam1 is the envelope's 45° camera.

Group	n	MAE	cMAE	GRF vert.	r R / L	Moments	EMG r
held out 7–11	396	5.13 (5.14)	3.02 (3.02)	16.28 (54.33)	0.77 / 0.79	2.41 (2.43)	0.24
training 2–5	319	5.62 (5.65)	3.54 (3.56)	16.15 (53.07)	0.78 / 0.77	2.42 (2.48)	0.22
walking	135	4.38 (4.40)	2.92 (2.93)	19.55 (76.11)	0.82 / 0.80	1.88 (1.79)	0.27
walking (trunk sway)	135	4.67 (4.67)	3.05 (3.05)	16.31 (84.14)	0.82 / 0.84	1.72 (1.53)	0.27
squats	87	5.79 (5.79)	3.26 (3.28)	6.32 (47.18)	0.53 / 0.47	1.58 (1.54)	0.17
sit-to-stand	90	5.39 (5.34)	2.92 (2.91)	11.58 (47.71)	0.68 / 0.80	1.55 (1.74)	0.20
drop jump	268	6.03 (6.03)	3.63 (3.63)	19.27 (32.21)	0.78 / 0.78	3.59 (3.73)	0.22
Cam0	144	5.21 (5.20)	3.14 (3.14)	14.95 (54.10)	0.75 / 0.83	2.44 (2.44)	0.23
Cam1 (45°)	141	5.62 (5.72)	3.33 (3.40)	15.50 (53.27)	0.73 / 0.81	2.54 (2.65)	0.22
Cam2	144	5.36 (5.36)	3.53 (3.53)	15.54 (52.72)	0.81 / 0.82	2.21 (2.31)	0.25
Cam3	143	5.35 (5.35)	3.20 (3.21)	15.96 (53.03)	0.81 / 0.70	2.44 (2.44)	0.23
Cam4	143	5.23 (5.23)	3.05 (3.05)	19.16 (55.68)	0.78 / 0.75	2.43 (2.45)	0.23

D.3 Ground reaction force

Table 13: Ground reaction force per activity, this run (previous run in brackets): MAE in % body weight on three axes with both feet summed, as OpenCap scored it; the share of the plate's stance frames the run's foot carries.

Activity	vertical	anterior	medio-lateral	3-axis mean	stance carried R / L
all	16.22 (53.77)	7.41 (6.45)	2.99 (3.19)	8.9 (21.1)	0.82 / 0.79
walking	19.55 (76.11)	12.03 (9.65)	3.02 (3.35)	11.5 (29.7)	0.82 / 0.75
walking (trunk sway)	16.31 (84.14)	10.86 (9.11)	3.49 (4.21)	10.2 (32.5)	0.83 / 0.85
squats	6.32 (47.18)	3.53 (2.56)	3.30 (4.17)	4.4 (18.0)	0.99 / 0.95
sit-to-stand	11.58 (47.71)	4.42 (4.74)	3.25 (3.51)	6.4 (18.7)	0.82 / 0.82
drop jump	19.27 (32.21)	5.61 (5.37)	2.54 (2.20)	9.1 (13.3)	0.76 / 0.71
OpenCap, 2 cameras	11.4	3.5	3.8	6.2	–

D.4 Residuals

Table 14: The pelvis residual this run leaves, per group: the median over trials of each trial's median force (% body weight) and moment (% weight × centre-of-mass height), the median of each trial's 95th-percentile force, and the mean share of frames within both limits (2% body weight, 1% weight × centre of mass height). No trial is within them on every frame.

Group	trials	force	moment	force, 95th pct.	frames within
all	715	7.0	8.90	24.4	21%
held out 7–11	396	7.1	9.31	23.9	21%
walking	135	7.8	10.99	23.0	6%
walking (trunk sway)	135	7.0	9.59	20.3	7%
squats	87	0.7	1.06	8.4	49%
sit-to-stand	90	0.2	2.82	8.8	46%
drop jump	268	10.9	9.32	85.5	17%

D.5 Muscle activations against EMG

Table 15: Each EMG channel against the chain's activation of that muscle, all 715 inputs: the share of frames on which both are on or both off, Pearson's r, and the lag (s) at the cross-correlation peak.

Muscle	on/off	r	lag	Muscle	on/off	r	lag
bflh l	0.65	0.13	-0.130	semiten r	0.70	0.23	-0.034
bflh r	0.65	0.13	-0.144	soleus l	0.70	0.26	-0.030
gasmed l	0.70	0.23	-0.009	soleus r	0.69	0.28	-0.041
gasmed r	0.69	0.26	-0.022	tibant l	0.59	0.22	+0.031
glmed1 l	0.51	0.13	+0.051	vaslat l	0.72	0.40	-0.003
glmed1 r	0.52	0.15	+0.040	vaslat r	0.72	0.42	-0.018
recfem l	0.56	-0.08	-0.002	vasmed l	0.72	0.36	-0.024
semiten l	0.66	0.18	-0.053	vasmed r	0.76	0.41	-0.033

Table 16: OpenCap's vasti symmetry index in squats: the AUC of the chain's index against the asymmetry the instructions asked for and against the EMG's own label (index over 1.15), previous run → this run. OpenCap: 0.83 against the EMG label.

Camera	AUC, instructed	AUC, EMG label
Cam0	0.47 → 0.84	0.64 → 0.75
Cam1	0.52 → 0.72	0.08 → 0.62
Cam2	0.60 → 0.95	0.52 → 0.84
Cam3	0.75 → 0.90	0.65 → 0.83
Cam4	0.53 → 0.81	0.25 → 0.48
all	0.56 → 0.73	0.45 → 0.66

D.6 Time per step

Table 17: The chain's wall seconds per step on one NVIDIA L4, over the 715 runs: median, range, and the median per frame (clips of 30 Hz).

Step	median s	min s	max s	s per frame
fit	21.825	14.483	623.317	0.303
set-up: physics model, GPU arrays, filtered path	4.697	3.878	32.851	0.061
footing	4.669	1.758	35.312	0.062
residual reduction, iterations	19.862	5.910	392.306	0.246
residual reduction, tracking and per-frame caps	4.603	1.714	45.888	0.061
checks: residual record, feet on the floor, tracking	2.120	1.111	17.879	0.028
static optimisation, stage 1 (ground reaction)	3.003	0.114	28.318	0.034
static optimisation, stage 2 (joint limits and muscles)	2.296	1.169	22.356	0.032
inverse dynamics, contacts and muscle paths	1.120	0.356	12.459	0.013
writing records and force measures	0.264	0.042	2.639	0.003
saving arrays	0.005	0.003	0.012	0.000
total	66.214	34.843	788.668	0.871

Table 18: Wall time per trial, the chain against published solves. OpenCap's solves are from its LabValidation release (80 muscles, windows of 0.99 to 1.68 s, solve only, hardware not recorded).

Chain	Clip	Wall time
OpenCap solve [15]	walk, drop jump, squat	4196, 4096, 2635 s
Moco, MocoTrack [12]	gait	about 130 min
OpenSim CMC [41]	gait	about 10 min
Warm start (chain), one L4	LabValidation, 715 trials	66 s median (35–789 s)

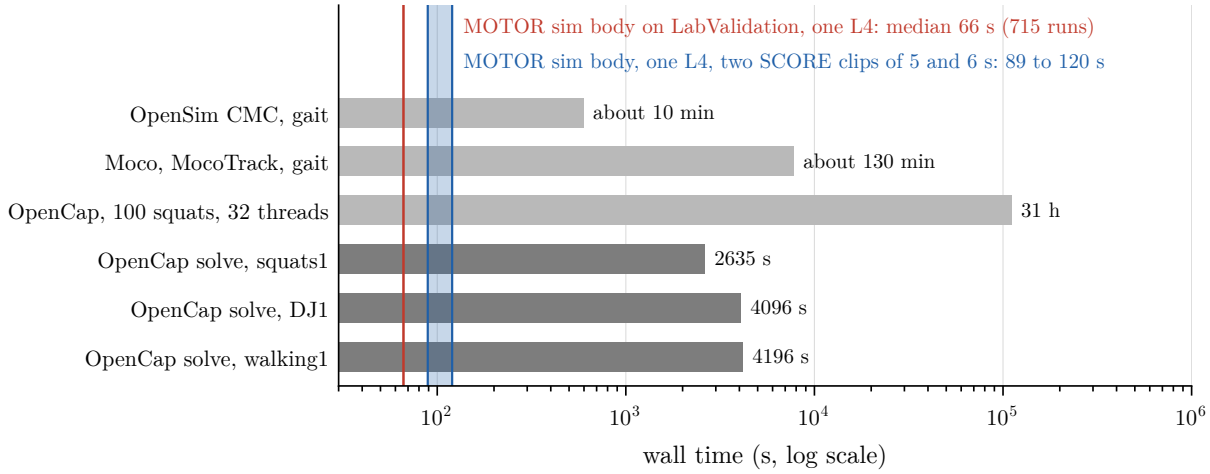


Figure 5: The per-frame chain's wall time per trial, not MOTOR's, on a log scale: OpenCap's recorded solves and published OpenSim times (grey), the chain on two SCORE clips (blue band) and its median on LabValidation (red). The legend's "MOTOR sim body" is the chain.

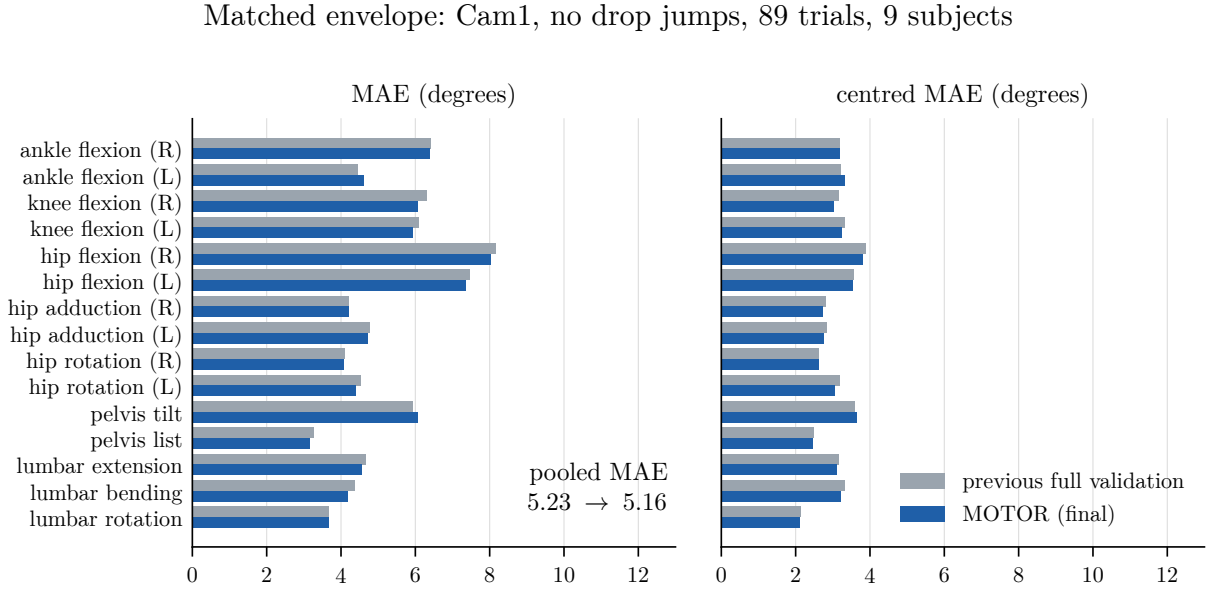


Figure 6: The per-frame chain's per-coordinate MAE and centred MAE on the matched set, its previous full run and this run; the chain's, not MOTOR's.

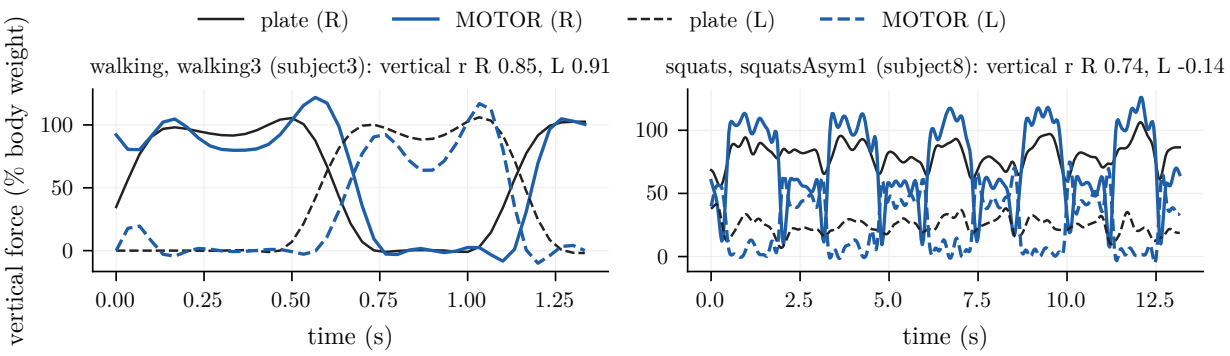


Figure 7: The per-frame chain's vertical force per foot against the plates, the 45° camera's median walking and squat trial, low-passed as OpenCap scores them. The legend's "MOTOR" is the chain, not MOTOR's whole-trial solve.

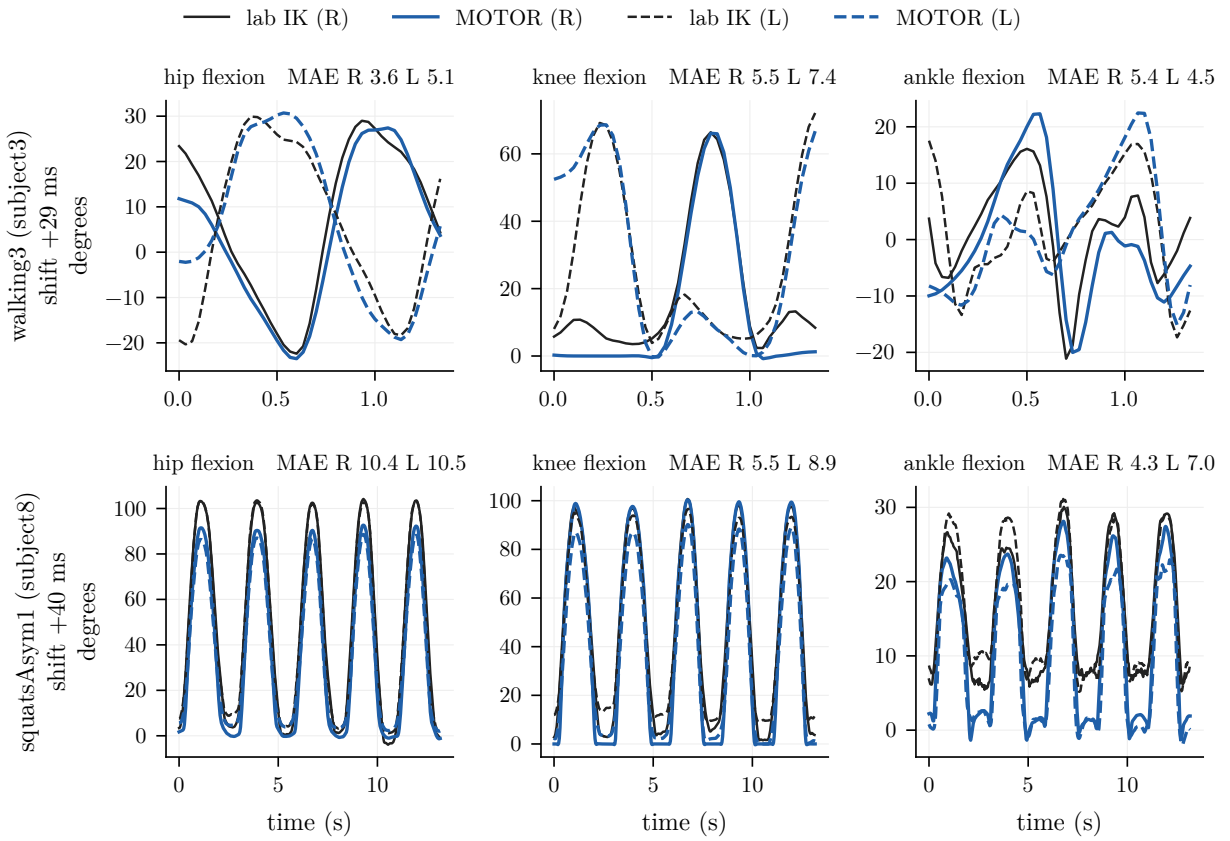


Figure 8: The per-frame chain's hip, knee and ankle flexion in the 45° camera's median walking and squat trial (by angle error) against the lab's inverse kinematics moved by the scored time shift. The legend's "MOTOR" is the chain.

D.7 What the chain shows

What the input carries. The chain follows its input, so the input's errors pass through. The elbows come out straighter than the lab's in every one of the 89 matched trials, by a mean 21.1° (right) and 23.7° (left), with centred errors of 3.0 and 4.6°. On 540 walking arm trials GEM-X's own elbow is a median 18.5° straighter than the markers', and the fit's elbow differs from GEM-X's by a median 2.5°. The pelvis tilt is offset in one direction in 82 of 89 matched trials, by a mean 5.3°. In the recorded runs, the knee of the leg far from the camera sits 37 to 44 mm from the centre of pressure in the frontal plane where the lab has 9 to 16 mm, which accounts for the error in the early-stance knee adduction moment.

Forces and residuals. The chain's pelvis residual stays above its limits (medians of 7.0% of body weight and 8.9% of weight times centre of mass height), its anterior ground reaction force error is about twice OpenCap's, and its activation pattern is partly off (rectus femoris does not follow its EMG, $r = -0.08$). Holding the residual within its limits is precisely what the whole-trial solve is designed to do.

Camera sampling and fast movements. GEM-X resamples every clip to 30 Hz, so an impact or a push-off is sampled at that rate whatever the phone recorded. Of the five activities the drop jumps have the chain's highest joint moment error (3.59% of weight times height, against 1.55 to 1.88 for the others), its largest pelvis residual (a median of 10.9% of body weight, its 95th percentile 85.5%), and two of the five inputs that failed did so because a drop jump frame's fit did not converge.

The fit's own records. The calibration (level 1) tunes on synthetic SOMA bodies, not on people. The last calibration run whose held-out score is recorded met its rule: its check bodies' score fell from 4.26 to 3.56 while the tuning bodies' fell from 4.53 to 3.44, and the rule's comparison, 3.56 against the tuning bodies' 3.51, was within 10%. The tuned point the fit uses now came from a later run with a broader objective, whose held-out score is not yet recorded. The bones are not yet all inside the skin: on the fifteen clips of the synthetic body of mean shape, 5.15% of the bone points lie outside it (averaged over body regions), against a target of under 5%.

D.8 Failures

Table 19: The 5 of 720 inputs without a score in the chain's run, by cause.

Cause	Inputs
The GPU fit did not stop in 3000 iterations (subject 10, squats1, two cameras)	2
A frame's fit did not converge in 300 steps (two drop jumps)	2
Passed its job's time limit (subject 10, squats1, a third camera)	1
Total	5

E Model and solver parameters

Table 20: The bone groups: each SOMA segment, the MyoFullBody bone that carries its rotation, and the bones the group covers. Every sided group exists on both sides; 7 middle and 7 sided groups make 21, of which 20 drive their frame bone.

SOMA segment	Frame bone	Bones in the group
Hips	pelvis	pelvis, sacrum
Spine1	lumbar 4	lumbar 5 and 4
Spine2	lumbar 1	lumbar 3 to 1
Chest	torso	torso, thorax, cervical spine
Neck1	neck	neck
Neck2	neck (drives nothing)	–
Head	head	head
Shoulder	scapula	clavicle, scapula
Arm	humerus	humerus
ForeArm	ulna	ulna, radius
Hand	lunate	carpals, metacarpals, phalanges
Leg	femur	femur
Shin	tibia	tibia
Foot	calcaneus	talus, calcaneus
ToeBase	toes	toes

Table 21: MOTOR's parameters in one place.

Part	Parameter	Value
Model	muscles, bones, joints	416, 86, 132
	joints: root, coupled, free	1, 36, 95
Landmarks	defined, used by the fit	102, 84
Fit	markers' goal λ_m	2.1169
	skin goal λ_s	10^3
	most steps per frame	300
Level 1	segment end's reach	40 mm
	held-out rule	check bodies within 10% of tuning bodies
Level 2	sampled frames	about 30
	landmark's and inside joint's reach	3 cm
Whole trial	spline	uniform cubic B-spline, one control point per frame
	smoothing cut-off	6 Hz
	contact band	2 cm over the floor
	foot stiffness	6×10^6 N/m
	dissipation, friction, slip transition	2.0 s/m, 0.8, 0.05 m/s
	effort share	0.1 of the fit's cost at the warm start
	residual limits	2% body weight; 1% weight × centre of mass height

F Disclosure, data and code

The code of MOTOR and of its validation was written by a coding agent (Claude Code) under the author's direction and review; this draft was written the same way. No result was hand-picked. The only settings chosen on LabValidation (the chain's three contact settings and box clearance, and the whole-trial solve's contact settings and effort share) were chosen on subjects 2 to 5 against their force plates, never against joint angles, and the tool that tunes MOTOR refuses subjects 7 to 11.

MOTOR, its validation tools and this paper's sources are open under the MIT licence at https:

//github.com/Babon-Innovations-b-v/score; every number in this paper is read from a file or record named in a comment beside it in the paper's source. We collected no new data from people. LabValidation is the public release of OpenCap's authors [15], used under its Apache-2.0 licence and fetched on demand, never redistributed.